

Aayushi Singh

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EDUCATION

University of Wisconsin - Madison

Master of Science in Electrical and Computer Engineering (Research)

Advisor: Prof. Joshua San Miguel

Current Coursework: Advanced Computer Architecture, High Performance Computing

University of Wisconsin - Madison

Bachelor of Science in Computer Engineering

GPA: 3.87/4.0 | Dean's List (7 semesters)

Coursework: Embedded Systems, Microprocessors, Digital Design, Computer Architecture, Robotics, Operating Systems

TECHNICAL SKILLS

Languages: C, SystemVerilog, Verilog, Python, LC-3 Assembly, Java, HTML, CSS, JavaScript

Computer Architecture: Cache Replacement Policies, Pipelined Processor Design, Memory Systems, Hazard Detection

Embedded Systems: ARM Cortex-M, PSoC6, MSP432, S32K344, FreeRTOS, AUTOSAR, Interrupts

Hardware Design: FPGA Design, Digital Logic, PCB Design, Processor Pipelines

Interfaces & Protocols: UART, SPI, I2C, Timers, Memory-Mapped I/O

Tools: ModelSim, Quartus, ARM Debug, Git, Linux, Docker, S32 Design Studio, FreeMASTER, HTCondor/CHTC

EXPERIENCE

Researcher

September 2025 – Present

Systems and Technologies Across the Computing Stack (STACS) Lab, UW-Madison

Energy-Aware Cache Replacement Policy | Advisor: Prof. Joshua San Miguel

- Building a cache simulator implementing LRU, FIFO, Belady's MIN, and an exhaustive oracle, and validating each policy against brute-force ground truth across exhaustively enumerated access sequences.
- Analyzing cache replacement behavior to quantify eviction overhead, finding that Belady's MIN achieves optimal hit rate but incurs 30% more evictions than necessary, with ~10% of sequences exhibiting avoidable churn.
- Designing an energy-aware replacement policy that directly targets eviction count, aiming to reduce cache churn while maintaining optimal hit rate.

Researcher

June 2024 – Present

Department of Electrical & Computer Engineering, UW-Madison

Energy-Efficient Inference for LLMs on Edge Hardware | Advisor: Prof. Setareh Behroozi

- Designing a selective bit-usage framework that splits model weights into fast-memory MSBs and on-demand LSBs, cutting memory and compute cost for on-device inference.
- Running saliency-guided bit-masking experiments on LLaMA-7B, showing saliency-guided masking substantially outperforms random masking as compression becomes more aggressive.
- Extending evaluation to accuracy benchmarks and conducting an oracle-gap analysis to quantify headroom for dynamic, per-token precision allocation.

Graduate Teaching Assistant

September 2026 – Present

Department of Electrical & Computer Engineering, UW-Madison

Digital System Fundamentals

- Leading Applied Homework lab sessions and guiding students through hands-on digital design assignments involving combinational/sequential logic, datapath design, FSMs, and basic processor architecture.
- Facilitating active-learning exercises on Boolean logic, datapaths and ALUs, sequential circuits, timing analysis, registers, control logic, and memory organization.
- Holding office hours and mentoring students through conceptual and implementation challenges, including debugging Verilog designs and interpreting ModelSim simulation waveforms.

Automotive Processor Field Application Engineer (FAE) Intern

May 2025 – August 2025

NXP Semiconductors

- Engineered a demo of NXP Safety Software on S32K344 MCU, demonstrating ISO 26262 ASIL compliance features.
- Built a custom FreeMASTER UI for real-time safety visualization, improving customer evaluation clarity.
- Validated safety and diagnostic features including BIST and eMCEM using targeted fault injection and debug tools.
- Reproduced customer-like workflows using ARM Debug, GCC, and AUTOSAR to accelerate issue isolation.

- Mentored students in digital logic, FSMs, datapath design, and basic processor architectures through hands-on debugging.
- Guided embedded development including C programming, interrupts, memory-mapped I/O, and peripheral interfaces.
- Supported debugging of timing, state behavior, and system integration issues using simulation waveforms.

PROJECTS

Runaway Alarm Robot

Best Capstone Project Award

- Built a self-balancing alarm robot that triggers motion on alarm events and navigates autonomously.
- Implemented PID-based balance and steering using IMU and Time-of-Flight sensor data for real-time control.
- Designed custom IMU, EEPROM, and magnetometer PCBs enabling full system integration and testing.

Pipelined Processor with Cache Hierarchy

- Built a 5-stage pipelined processor with branch flushing, hazard detection, and data forwarding for a 16-bit ISA.
- Designed 2-way set-associative I/D caches with LRU, write-through and write-allocate policies.
- Developed directed testbenches in ModelSim to validate control hazards, data hazards, and cache miss-handling FSM behavior.

Autonomous Robot on FPGA

- Built an FPGA-based autonomous Knight's Tour robot with PID control and SPI sensor feedback.
- Reduced synthesized area by 12.5 percent while resolving timing and hold-time violations.
- Built functional verification testbenches for navigation and motor-control and debugged failures using simulation waveforms.

LEADERSHIP

Vice President of Operations

September 2024 – May 2026

Women in Electrical & Computer Engineering (WECE)

- Designed the organization website, maintaining pages for leadership, events, membership, and organizational resources.
- Managed internal operations across executive teams, including scheduling, documentation, communications, and resources.
- Led merchandise operations and provided direction to Social Media & Marketing and New Member Directors on outreach and content execution.